Brain Research, 627 (1993) 25–33
© 1993 Elsevier Science Publishers B.V. All rights reserved 0006-8993/93/$06.00

BRES 19359

The use of the optical disector to estimate the number of neurons, glial and endothelial cells in the spinal cord of the mouse – with a comparative note on the rat spinal cord

Roger Bjugn *

Department of Anatomy and Cell Biology, University of Bergen, Årstadveien 19, N-5009 Bergen, Norway and Stereological Research Laboratory, Århus University, DK-8000 Århus C, Denmark

(Accepted 8 June 1993)

Key words: Cell count; Central nervous system; Mammal; Neuroglia; Quantification; Stereology

By means of stereological techniques such as the optical disector, the total number of cells in the spinal cords of five albino mice were estimated. The mean number of cells in the gray and the white matter was 9.0 and 4.1 million, respectively. Of the cells in the gray matter, 4.0 million were classified as neurons, 2.8 million as glial and 1.7 million as endothelial cells whereas 0.4 million remained unclassified. Of the neurons in the gray matter, 1.2 million were located in the cervical region, 1.4 million in the thoracic region, 0.9 million in the lumbar region and 0.4 million in the sacro-coccygeal region. The gray matter in the spinal cord of the mouse contained relatively more neurons and less glial cells than the gray matter in the rat spinal cord. The fraction of cells in the cervical, thoracic, lumbar and sacro-coccygeal part of the cord were, however, quite similar in the two species. In both species the number of neurons in the ventral horns made up about 8–9% of the total neuron population in the gray matter of the cord.

INTRODUCTION

Quantification is of uttermost importance both in pure descriptive and experimental settings. The complexity of the central nervous system has until the last few years made it impossible to get unbiased estimates of cell number in this region. The stereological tools recently developed have, however, changed this (for review see refs. 14 and 15). Reliable data can now be obtained quite easily[3,5,12,17,22,23,32,33]. Stereological techniques have also been used in combination with immunocytochemical methods[25]. Although an absolute number in itself may be of minor importance, a reliable estimate makes it possible to make qualified comparisons between different species, between different regions of the central nervous system, during development, in diseases and in experimental settings. In that way, our understanding of the organization of the central nervous system, or any other system, can be facilitated.

The albino mouse is one of the most frequently used animals in biological research. However, data on cell number in its central nervous system are only fragmentary. Regarding the spinal cord, only two studies on cell number have, to my knowledge, been published[26,27]. Both papers deal with the cervical part of the cord. The aims of the present investigation were to estimate the total number of neurons, glial and endothelial cells in the entire spinal cord of the mouse, to obtain an estimate of how the cells are distributed within the cord, and to make some comparative notes on the spinal cord of the rat.

MATERIALS AND METHODS

The technical procedures employed were generally those described in a previous study on the rat spinal cord[3]. However, as cell classification and practical procedures were somewhat modified, an outline of the methods used is given below.

Tissue preparation

Five adult albino male mice (Bom: NMRI) were used (Table I). Anesthesia was obtained by a subcutaneous injection of midazolam (6.25 mg/kg b.wt.) and fentanyl (0.25 mg/kg b.wt.)/fluanosin (12.5 mg/kg b.wt.). Transcardiac vascular perfusion fixation was done by

* Corresponding author. Fax: (47) 5-206360.

means of a peristaltic pump (40 ml/min for 10 min), using 3% glutaraldehyde in a 0.1 M Sörensen phosphate-buffer (prepared from $Na_2HPO_4 \cdot 2H_2O$ and $NaH_2PO_4 \cdot H_2O$, ref. 10) containing 0.08 M sucrose (pH 7.2, 23–25°C).

The spinal cords were isolated and their lengths measured (Table I). The rostral limit of the cord was set approximately 0.5 mm below the upper border of the atlas while the caudal limit was set at the level where the conus medullaris tapered off into an even-sized filum terminale. Each cord was transected into 8 slabs of equal length. All slabs were rinsed in the fixative vehicle. By means of a razorblade, one transverse slice and one oblique slice were then cut from all slabs (Fig. 1), except from the most caudal slab where only a transverse slice was cut. The transverse slices were used for the estimation of the spinal cord volume (see Stereological techniques), and they were always cut from the rostral end of the slabs. The oblique slices were used for the estimation of the numerical densities of cells (see Stereological techniques), and they were cut as depicted in Fig. 1 and always along the ventral-dorsal axis of the cord.

All slices were dehydrated in increasing concentrations of ethanol (70–90–99%) and infiltrated, and finally embedded in 2-hydroxyethyl methacrylate. From all embedded slices, 35-μm-thick sections were cut, stretched on a water bath, and stained by a Giemsa technique (see Table 2 in ref. 5).

Stereology

Cell classification

Based on the assumption that all cells had one, and only one nucleus, the estimates of cell number were based on counting nuclei. The cells were classified either as neurons, glial cells, endothelial cells (including pericytes and smooth muscle cells), or as unidentified cells. Cells classified as neurons were subdivided into three categories: (1) those with a prominent nucleolus and a clearly defined cytoplasm containing distinct Nissl bodies (Fig. 2A); (2) those with a prominent nucleolus and a clearly defined cytoplasm without distinct Nissl bodies (Fig. 2B); (3) those with nuclei having marked indentations in the nuclear membrane and only scant or no apparent cytoplasm. The majority of such cells contained a distinct nucleolus and were predominantly located in the dorsal horns (Fig. 2C).

Most neurons appeared with a rather pale nucleus compared to that of endothelial and glial cells. Endothelial cells were defined by their location around a blood vessel or by the curved shape of their nuclei (Fig. 2B). Cells appearing without cytoplasm and without marked indentations in the nuclear membrane, and not classified as endothelial cells, were classified as glial cells (Fig. 2A,B,D). The remaining cells were classified as 'unidentified' (Fig. 2E).

Stereological techniques

The areas of gray and white matter on each cross section (taken from the rostral part of each slab, see Tissue preparation) were estimated by point counting techniques[3]. The volume of each slab, except the most caudal one, was estimated by the following formula:

$$V_i = \frac{A_i + A_{i+1}}{2} \cdot h$$

TABLE I

Data on the five mice used and their spinal cords

CV = observed coefficient of variation. SD = standard deviation.

Animal number	*Animal weight (g)*	*Length of spinal cord (mm)*	*Volume (mm^3)*	
			Gray matter	*White matter*
1	33	45	25.8	19.3
2	28	44	25.2	22.2
3	30	44	24.1	21.0
4	30	43	22.9	20.8
5	37	43	24.4	23.6
Mean	31.6	43.8	24.48	21.38
SD	3.5	0.8	1.11	1.61
CV	0.111	0.018	0.045	0.075

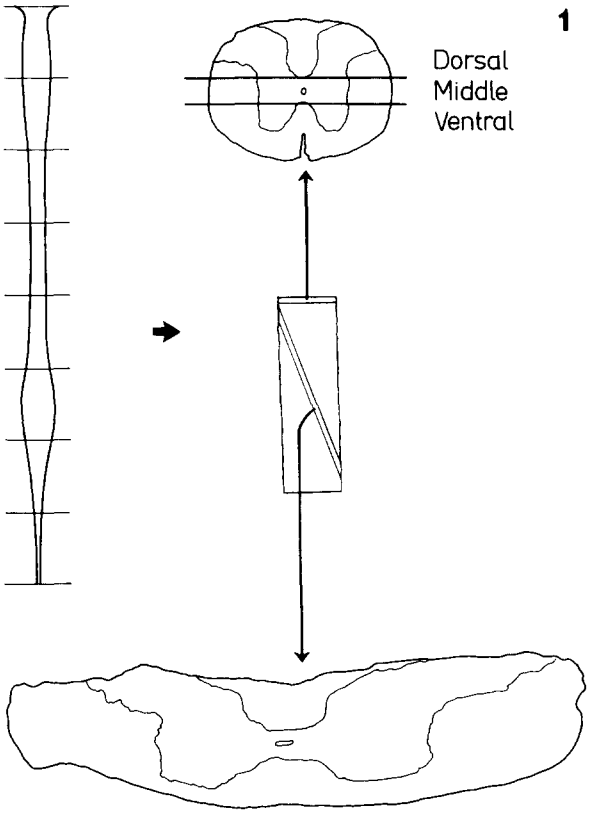


Fig. 1. Figure showing how the spinal cord of each mouse was cut into eigth slabs of equal length, how one transverse and one oblique slice was cut from each slab and how the light microscopical sections from the transverse and oblique slices appeared. The partition between the ventral horns, the middle portion and the dorsal horns is depicted on the transverse section.

where V_i denotes volume of slab i, $i = 1 \ldots 8$; A_i denotes the rostral cross sectional area of slab i; and h denotes the length of the slabs (all slabs from one particular cord were of equal length).

Since the caudal cross sectional area for the eighth, and most caudal, slab was not measured, the volume of the eighth slab was estimated using the following formula:

$$V_8 = A_8 \cdot h$$

The total volume (reference volume) of gray and white matter in the spinal cord could then be estimated from the following equation:

$$V = \frac{A_1 + A_2}{2} \cdot h + \frac{A_2 + A_3}{2} \cdot h + \ldots + \frac{A_7 + A_8}{2} \cdot h + A_8 \cdot h$$

$$= \left(\frac{A_1}{2} + A_2 + \ldots + A_7 + \frac{3 \cdot A_8}{2} \right) \cdot h$$

The number of cells present in a given tissue volume is called the numerical density (N_V). In the present investigation, the numerical densities of various cell categories in gray and white matter were estimated on the oblique sections by means of an optical disector[13,14,30]. The counting frame area of the optical disector was 216 μm^2 and the depth 15 μm (measured by a microcator). The use of oblique sections of the spinal cord made it possible to disregard possible cyclic variations in numerical densities of cells along the cord. The 2-dimensional sampling density for the systematically random sampling scheme was approximately 16 visual fields per mm^2. An estimate of the total number of cells in gray and white matter was obtained by multiplying the appropriate N_V with the corresponding reference volume.

In one animal (animal 2), the numerical densities and reference volumes were recorded separately for the ventral horns, the middle portion and the dorsal horns, respectively (see Fig. 1 for partition).

Equipment

The equipment used are described in a previously published paper[3]. The only difference was that just one counting frame (with an area of 216 μm^2, at the tissue level) was used in the present investigation.

Statistical analysis

The biological variation (coefficient of variation; BCV) of cell number between animals was estimated from the calculated stereological sampling variation (coefficient of error: CE = SEM/mean) and the observed total variation in cell number (coefficient of variation: CV = SD/mean) using the relationship; $CV^2 = BCV^2 + CE^2$. The coefficient of error for the estimate of cell number for each animal was calculated as shown by West and Gundersen[32], an example is given in Table III (see also ref. 3).

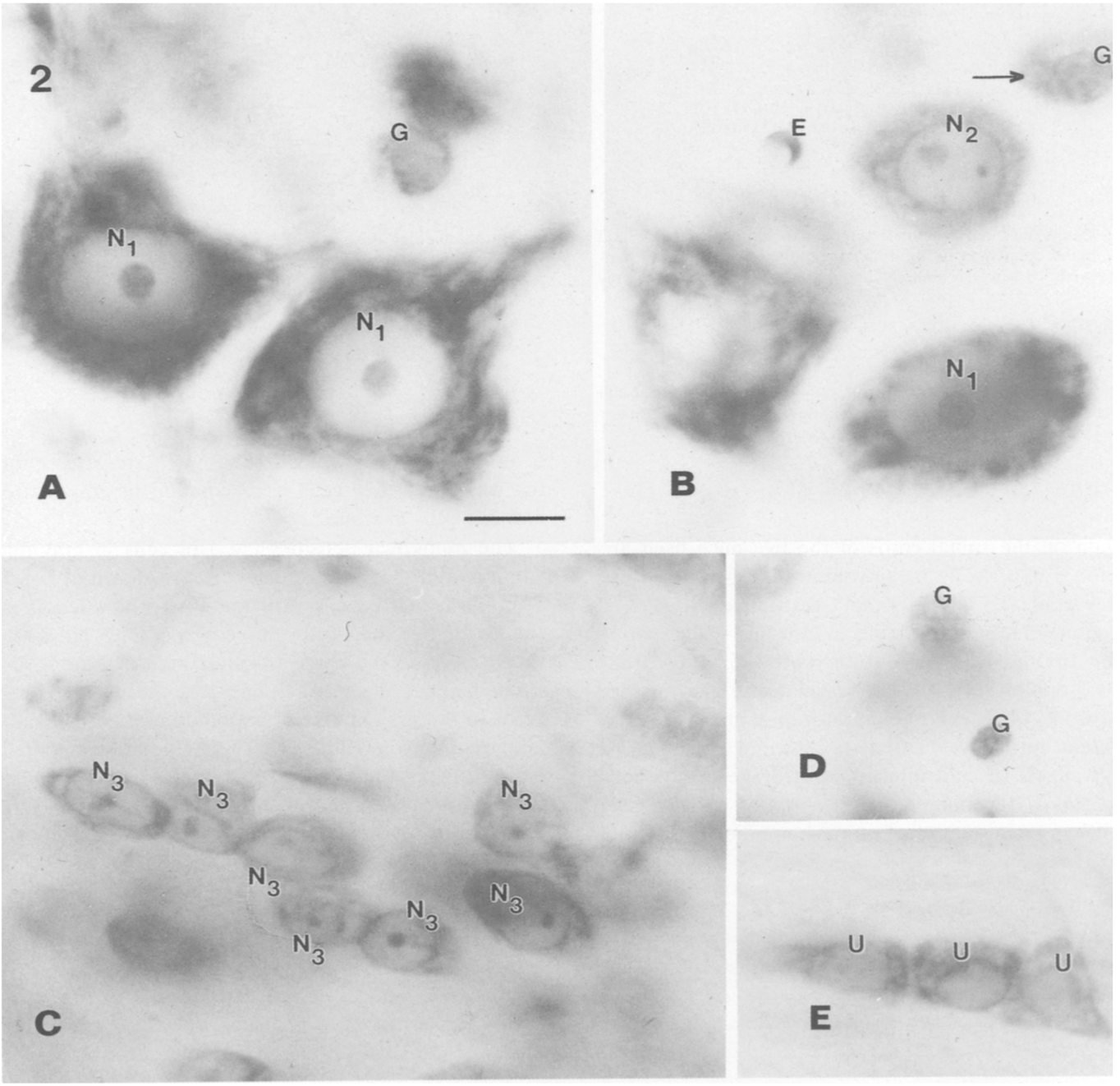


Fig. 2. Light micrographs of neurons (N), endothelial (E), glial (G) and unidentified (U) cells in the spinal cord of the mouse. Cells believed to be neurons were subdivided into three categories (compare Table V): The first category included those with a prominent nucleolus and a clearly defined cytoplasm containing distinct Nissl bodies (N_1). The second category included those with a prominent nucleolus and a clearly defined cytoplasm without distinct Nissl bodies (N_2). The third category included those with nuclei having marked indentations in the nuclear membrane and only scant or no apparent cytoplasm (N_3). The majority of cells in the latter category contained a nucleolus and were predominantly located in the dorsal horns. A,B,D: from gray matter in the ventral horns. C: from gray matter in the dorsal horns. E: from white matter. Arrow in B points to glial cell nucleus beneath the labelled cell nucleus. Photographs of 35-μm-thick 2-hydroxyethyl-methacrylate sections. Bar = 10 μm.

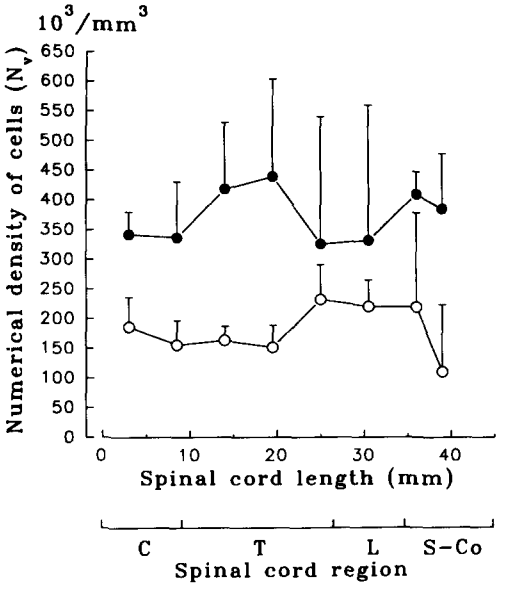


Fig. 3. Number of cells per mm^3 (numerical density; N_V) gray and white matter related to position along the mouse spinal cord. Standard deviations between animals are shown by vertical bars. •, gray matter; ○, white matter.

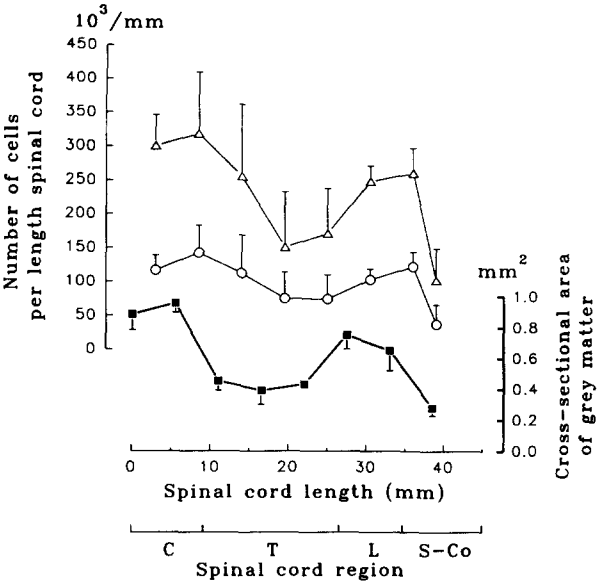


Fig. 4. Number of all cells and neurons specifically per mm length (N_L) of gray matter related to position along the spinal cord of the mouse. N_L = (area of cross-section)·N_V (numerical density). Cross-sectional area of gray matter along the cord is also given. Standard deviations are shown by vertical bars. △, all cells; ○, neurons; ■, cross-sectional area.

Comparative notes

Data from an earlier study on the number of cells in the spinal cord of the rat[3] were used for the comparative notes on cell number in the spinal cord of the mouse and the rat.

RESULTS

Data on the spinal cords of the five mice used are given in Table I. The mean volumes of gray and white matter were 24.5 and 21.4 mm^3, respectively.

The mean numbers of cells in gray and white matter of the five mouse spinal cords were 9.0 (CV = 0.08) and 4.1 million (CV = 0.16), respectively. The number of neurons, glial, endothelial and unidentified cells in the spinal cords investigated are shown in Table II. For the estimates of cell number, an average of 201 cells (169–234) were counted in gray matter and 80 cells (69–96) in white matter of animals number 1, 3, 4 and 5. In animal number 2, where a more extensive counting was made, 620 cells were counted in gray matter and 245 cells in white.

The numerical densities (N_V) of cells in gray and white matter along the spinal cord are shown in Fig. 3. The number of cells in gray matter per mm length of the cord, together with the coresponding cross-sectional areas of the cord, are illustrated in Fig. 4. The mean number of cells belonging to the different cell types in the gray matter of various spinal cord regions are presented in Table IV. The estimated number of neurons, glial, endothelial and unidentified cells located in the ventral horns, the middle portion and the dorsal horns in animal number 2 are given in Table V.

TABLE II

Number of neurons (N), glial (G), endothelial (E) and unidentified (U) cells (10^6) in gray and white matter of five mouse spinal cords

CV = observed coefficient of variation. SD = standard deviation. Σ = total cell number.

Animal number	*Gray matter*					*White matter*				
	Σ	*G*	*E*	*N*	*U*	Σ	*G*	*E*	*N*	*U*
1	8.6	2.4	1.7	4.2	0.3	3.6	1.3	0.9	0.6	0.8
2	8.9	3.2	1.7	3.8	0.2	3.5	2.4	0.6	0.1	0.4
3	9.3	2.7	1.7	4.5	0.4	3.9	2.3	0.8	0.2	0.6
4	10.0	2.9	2.3	4.3	0.5	4.6	2.2	1.2	0.7	0.5
5	8.0	2.6	1.2	3.4	0.8	5.0	3.7	0.4	0.2	0.7
Mean	8.96	2.76	1.72	4.04	0.44	4.12	2.38	0.78	0.36	0.60
SD	0.75	0.30	0.39	0.44	0.23	0.65	0.86	0.30	0.27	0.16
CV	0.084	0.109	0.227	0.109	0.523	0.158	0.361	0.385	0.750	0.267

TABLE III

Data from animal 1 showing how the coefficient of error (CE) for the estimate of cell number in gray matter was calculated

F = number of fields where cells were counted. P = number of points falling on the transverse sections. Q^- = number of cells counted in the oblique sections.

Slab	Q^-	F	P	$X_i = Q^- \cdot P/F$	$A = X_i \cdot X_i$	$B = X_i \cdot X_{i+1}$	$C = X_i \cdot X_{i+2}$
1	39	36	23	24.9	620	647	281
2	34	34	26	26.0	676	294	200
3	17	15	10	11.3	128	87	158
4	17	22	10	7.7	59	108	179
5	52	41	11	14.0	196	326	218
6	37	35	22	23.3	543	363	112
7	10	9	14	15.6	243	75	–
8	3	5	8	4.8	23	–	–
Σ	$Q^- = 209$	$F = 197$	$P = 124$	$X_i = 127.6$	$A = 2{,}488$	$B = 1{,}900$	$C = 1{,}148$

$$CE = \frac{\sqrt{\frac{3A + C - 4B}{12}}}{\Sigma X_i} = \frac{\sqrt{84.3}}{127.6} = 0.072$$

Note that

$$\frac{\Sigma Q^-}{\Sigma F} \cdot \Sigma P = \frac{209}{197} \cdot 124 = 131.6 \neq \Sigma\left(\frac{Q^-}{F} \cdot P\right) = 127.6$$

The latter is used only for estimating CE, while the first is used for the unbiased estimate of cell number (see ref. 32).

DISCUSSION

Biology

Cell classification

A detailed account on criteria for cell classification is given in a previous paper on the spinal cord of the rat[3] and only a brief resumé is given here.

No clear-cut criteria for neuron identification could be found in the present study. That cells with a prominent nucleolus and a clearly defined cytoplasm containing distinct Nissl bodies (Fig. 2A) are neurons, is probably uncontroversial. Of the cells classified as neurons in the ventral horns, 95% fulfilled these criteria. However, in the dorsal horns only 16% of the cells classified as neurons met the above-mentioned criteria.

TABLE IV

The mean number (10^6) of neurons, glial, endothelial and unidentified cells in gray matter of various regions of the mouse spinal cord

The mean volumes (mm^3) of each region are also presented.

Region	*Volume*	*Cell Category*				
		Glial	*Endothelial*	*Neurons*	*Unidentified*	*Total cell number*
Cervical	7.4	0.8	0.5	1.2	0.1	2.6
Thoracic	8.3	0.9	0.6	1.4	0.1	3.0
Lumbal	5.7	0.6	0.4	0.9	0.1	2.0
Sacro-coccygeal	3.8	0.4	0.3	0.6	0.1	1.4

The second category of cells classified as neurons included those with a prominent nucleolus and a clearly defined cytoplasm without distinct Nissl bodies (Fig. 2B). 'Distinct' Nissl bodies are dependent on the staining technique used, but a distinct nucleolus in a pale nucleus usually defines a neuron[24]. The third category of cells classified as neurons included those with only scant or no apparent cytoplasm and with nuclei having

TABLE V

Volumes (mm^3) of the ventral horns, the middle portion and the dorsal horns, together with the corresponding number of cells (10^6), in gray matter of one mouse spinal cord (animal 2) down to the upper sacral segments

This part contained 94% of gray matter in this particular cord. Percentage of neurons fulfilling which criteria is also given.

	Ventral	*Middle*	*Dorsal*
Volume	6.6	6.2	9.5
Number of glia	1.2	0.7	1.0
Number of endothelia	0.5	0.5	0.6
Number of unidentified	0	0	0.1
Number of neurons	0.3	0.6	2.3
Category 1 [a]	95%	54%	16%
Category 2 [b]	5%	25%	44%
Category 3 [c]	0%	21%	40%

[a] Cells having a prominent nucleolus and a clearly defined cytoplasm containing distinct Nissl bodies. [b] Cells having a prominent nucleolus and a clearly defined cytoplasm without distinct Nissl bodies. [c] Cells having nuclei with marked invaginations in the nuclear membrane and only scant or no apparent cytoplasm. The majority of such cells contained a nucleolus and were predominantly located in the dorsal horns.

marked indentations in the nuclear membrane (Fig. 2C). According to electron microscopic[20] and immuno-electron microscopic[31] investigations, neurons in the dorsal horns of the rat are characterized by these features. By transmission electron microscopy we have observed that cells located in the dorsal horns of the mouse spinal cord, and with the above-mentioned characteristics, may have axo-somatic synapses. It therefore seems reasonable to classify such cells as neurons.

Neurons not having the morphological criteria mentioned above will in the present study have been classified either as glial or unidentified cells. Glial cells may have been included in both the second and third neuron category. Due to this problem of classification, the estimates given for the number of cells in each cell category should be considered with care. The estimate of the total number of cells will, however, not be affected. Combining sections for transmission electron microscopy with a physical disector (see ref. 7), would probably facilitate cell classification. Such a procedure is, however, rather time-consuming.

The dorsal horns of the grey matter contain projection (or tract) neurons, which convey sensory information to supraspinal parts of the central nervous system, and interneurons. The middle (or intermediate) zone contains projection neurons, which transmit sensory information to the cerebellum, interneurons and preganglionic autonomic neurons. The ventral horns contain motoneurons, which control the limbs and the trunk, and interneurons[19].

The term 'interneuron' is usually applied to neurons which have axonal projections entirely within the spinal cord. Projection neurons may, however, have collaterals that end within the cord[8]. Of the neurons in the spinal cord, the vast majority are interneurons[8,35]. According to Chung et al.[9], about 97% of the neurons in the S2 segment of the rat spinal cord are interneurons, approximately 1% are projection neurons and 1–2% are motoneurons. Although these figures most probably are not representative for the entire spinal cord, they indicate that data processing within the spinal cord is a major task and involves most of the neurons.

The present estimates of the number of neurons in the gray matter of the mouse spinal cord include all kinds of neurons. It was not attempted to differentiate between projection neurons, interneurons, preganglionic autonomic, or motoneurons, and it is in my opinion, futile with the staining and identification techniques available at present. However, as the dorsal horns contained the majority of neurons in the gray matter (more than 70%, see Table V), neurons related to sensory processing are the most numerous.

Previous studies on the number of cells in the mouse spinal cord

The estimated volume of gray matter in the cervical region was 7.4 mm^3, and the number of neurons and glial cells were 1.2 and 0.8 million, respectively. In 120-day-old mice, Sakla[26,27] found the cervical gray matter volume to be 8.1 mm^3 and the number of both neurons and glial cells to be 1.4 million each. The difference between the present and Saklas' estimates of cell number is most likely due to the different cell-counting techniques used. Biological variation between different mouse strains may also have contributed to the difference.

Comparative notes on the cell number and the cell distribution

The overall brain-body relation in mammalia is often expressed as $E = kP^b$, where E is brain mass, k is a constant, P is body mass and b is a constant in the range of 0.67 to 0.75. Although the general trend for mammalia is like this, the clear-cut relation breaks down when comparing small number of species or when comparing different animals within one species[18,34]. The relation between neuron number and body mass is even more muddled; partly because reliable data are limited and partly because neuron number shows such diversity between individuals of the same species and between individuals of different species but of similar body weight[34]. However, despite these difficulties, and the problems concerned with cell identification, some comparative notes on the spinal cord of the mouse and the rat can be made using our data on cell number and cell distribution.

The estimates of cell number in the spinal cord of the mouse were from males whereas the estimates from the rat spinal cord were from females. It is known that sexual dimorphism exists in the central nervous system of mammalia. Regarding the spinal cord, the spinal nucleus of the bulbocavernosus[6,11] contains some hundred neurons more in the adult male than in the adult female. Apart from this, I am unaware of other sexual differences in the spinal cord. Until fully investigated, all statements on cell number and distribution related to sex should, however, be with this restraint in mind.

There is about a 10-fold weight difference and a 4.5-fold surface area difference (surface area $\approx 10 \cdot$ body mass$^{0.67}$, see ref. 29) between the rat and the mouse. According to our data, there is a 2.8-times difference between overall cell number in the spinal cord of the rat[3] (36.2 million) and the mouse (13.1 million). The corresponding ratios between the number

of cells in the gray and the white matter were 2.3 (21.1 vs. 9.0 million) and 3.7 (15.1 vs. 4.1 million), respectively. Ependymal cells are not included in these number estimates. However, as there are just about 80,000 ependymal cells in the mouse spinal cord[1,2] and about 320,000 in the rat spinal cord (unpublished observations, R. Bjugn), the estimates of the total cell number will only marginally be affected.

In the mouse, the numerical density (N_V) of all cells was 1.9 times higher in the gray matter than in the white. In the rat, the corresponding ratio was 2.3[3]. In the gray matter, the N_V of both all cells, and neurons specifically, were substantially higher in the mouse than in the rat. This is in agreement with previous studies on the cerebral cortex showing that the N_V generally is in inverse proportion to species size[4].

Fig. 5 shows that larger species (the rat) has relatively more glial cells and less neurons in the gray matter of the spinal cord. The glial index (N_N = number of glial cells/number of neurons) in the gray matter was 1.6 in the rat[3] and 0.7 in the mouse. How the glial index is related to phylogenesis or species size is at present not fully known. However, the glial index generally appears to be higher the larger the species size[4].

The fractions of cells located in the cervical, thoracic, lumbar and sacro-coccygeal part of the cord were quite similar in the two species (Fig. 6). The fractions of cells located in the ventral horns, the middle portion and the dorsal horns appeared to be similar too. The latter data are based, however, on estimates from one animal from each species only.

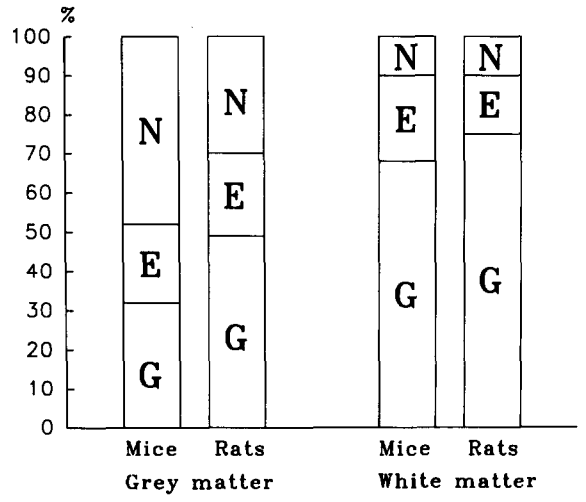


Fig. 5. Percentages of neurons (N), endothelial (E) and glial (G) cells in the gray and the white matter of the spinal cord of the mouse and the rat. The unidentified cells in the spinal cord of the mouse have been excluded in this figure (see Table II).

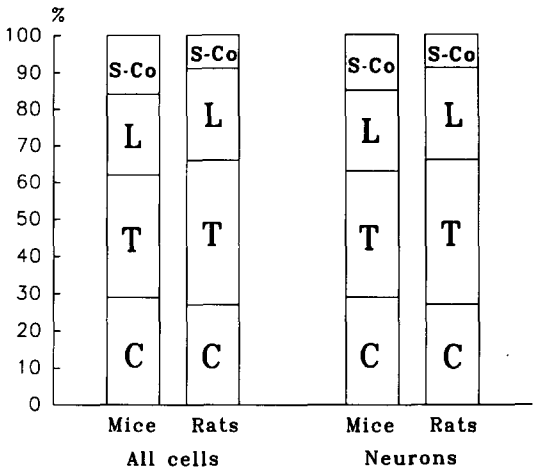


Fig. 6. Percentages of all cells, and neurons specifically, in gray matter of the cervical (C), thoracic (T), lumbar (L) and sacro-coccygeal (Sa-C) segments of the spinal cord of the mouse and the rat.

The gray matter in the mouse spinal cord contained, on the average, 4.0 million neurons compared to 6.4 million in the rat[3]. In both species, ventral horn neurons made up about 8–9% of the entire neuron population in the gray matter (about 0.6 millions in the rat and 0.3 millions in the mouse). These estimates of number of ventral horn neurons include all kinds of neurons, not only the motoneurons. The ratios between α-, β- and τ-motoneurons and interneurons in the ventral horns are, to my knowledge, not known. Regarding the motoneurons, it has, however, been stated that 'motoneurons in the gamma size range' constitute about 30% of the horseradish peroxidase-labelled motoneurons related to the medial gastrocnemius muscle[16]. Data on muscle fibres and motoneurons indicate that the larger the species, the larger is the number of muscle fibres present in a given muscle, and more fibres are innervated by each motoneuron[28]. One should bear in mind, however, that not all motoneurons are confined to the ventral horns. Retrograde labelling by horseradish peroxidase have localized motoneurons in the middle portion of gray matter and occasionally even in the most basal parts of the dorsal horns[21]. Whether the fraction of motoneurons is constant in different mammalian species awaits further studies.

Tissue selection and treatment

A discussion on the rostral and caudal limits of the spinal cord and the tissue treatment has been given elsewhere[3]. I will, however, point out that no correction for volumetric changes during tissue processing has been made in the present estimates.

Sampling scheme, stereological techniques and amount of work

Based on observations in the rat spinal cord the number of cells counted in the present study, about 200 in gray and 80 in white matter, was selected in order to give a CE of about 0.1 on the number estimates for the different cell categories in gray matter and a CE of 0.1 for the total cell number in white matter.

The calculated CE's for number estimates in gray matter were in the range of 0.06–0.20 ($CE \approx 1/\sqrt{N}$, where N is the number of cells belonging to one cell type counted in one spinal cord). The uncertainty in number estimates for one particular cell type is, however, not confined to the number of cells counted, it is cell identification that is the major problem. The observed coefficient of variation (CV) for total cell number in gray matter was 0.08 and the mean coefficient of error (CE) 0.07 (0.06–0.09). The biological variance $[(BCV/CV)^2]$ accordingly made up 25% of the observed variance for total cell number. In white matter, with a CV of 0.16 and a mean CE of 0.07, the biological variance made up 72% of the observed variance.

Acknowledgements. The technical assistance of Anette Funder, Lise Gjerrild, Maj-Britt Lundorf and Lone Melgaard at the Stereological Research Laboratory is greatly appreciated. Bico, Denmark is thanked for their donation of technical equipment. Financial support to R.B. was given by the Nordic Council of Ministers.

REFERENCES

1 Bjugn, R., Bøe, R. and Haugland, H.K., A stereological study of the ependyma of the mouse spinal cord. With a comparative note on the choroid plexus ependyma, *J. Anat.*, 166 (1989) 171–178.
2 Bjugn, R., Bøe, R. and Haugland, H.K., Erratum note. A stereological study of the ependyma of the mouse spinal cord. With a comparative note on the choroid plexus ependyma, *J. Anat.*, 168 (1990) 279.
3 Bjugn, R. and Gundersen, H.J.G., Estimate of the total number of neurons and glial and endothelial cells in the rat spinal cord by means of the optical disector, *J. Comp. Neurol.*, 328 (1993) 406–414.
4 Blinkov, S.M. and Glezer, I.I., *The Human Brain in Figures and Tables. A Quantitative Handbook*, Plenum, New York, 1968, pp. 1–482.
5 Braendgaard, H., Evans, S.M., Howard, C.V. and Gundersen, H.J.G., The total number of neurons in the human neocortex unbiasedly estimated using optical disectors, *J. Microsc.*, 157 (1990) 285–304.
6 Breedlove, S.M., Steroid influences on the development and function of a neuromuscular system, *Prog. Brain Res.*, 61 (1984) 147–170.
7 Brændgaard, H. and Gundersen, H.J.G., The impact of recent stereological advances on quantitative studies of the nervous system, *J. Neurosci. Methods*, 18 (1986) 39–78.
8 Burke, R.E. and Rudomin, P., Spinal neurons and synapses. In E.R. Kandel (Ed.), *Handbook of Physiology*, Section 1, *The Nervous System*, Vol. I, *Cellular Biology of Neurons*, Part 2, American Physiological Society, Bethesda, MD, 1977, pp. 877–944.
9 Chung, K., Kevetter, G.A., Willis, W.D. and Coggeshall, R.E., An estimate of the ratio of propriospinal to long tract neurons in the sacral spinal cord of the rat, *Neurosci. Lett.*, 44 (1984) 173–177.
10 Dawson, R.M.C., Elliot, D.C., Elliot, W.H. and Jones, K.M., pH, buffers, and physiological media. In: *Data for Biochemical Research*, Oxford University Press, Oxford, 1986, pp. 417–448.
11 DeVoogd, T.J., Androgens can affect the morphology of mammalian CNS neurons in adulthood, *Trends Neurosci.*, 10 (1987) 341–342.
12 Evans, S.M., *Stereological Investigation of the Ageing of Human Cerebral Neocortex (Thesis)*, University of Liverpool, Liverpool, 1990.
13 Gundersen, H.J.G., Stereology of arbitrary particles. A review of unbiased number and size estimators and the presentation of some new ones, in memory of William R. Thompson, *J. Microsc.*, 143 (1986) 3–45.
14 Gundersen, H.J.G., Bagger, P., Bendtsen, T.F., Evans, S.M., Korbo, L., Marcussen, N., Møller, A., Nielsen, K., Nyengaard, J.R., Pakkenberg, B., Sørensen, F.B., Vesterby, A. and West, M.J., The new stereological tools: disector, fractionator, nucleator and point sampled intercepts and their use in pathological research and diagnosis, *APMIS*, 96 (1988) 857–881.
15 Gundersen, H.J.G., Bendtsen, T.F., Korbo, L., Marcussen, N., Møller, A., Nielsen, K., Nyengaard, J.R., Pakkenberg, B., Sørensen, F.B., Vesterby, A. and West, M.J., Some new, simple and efficient stereological methods and their use in pathological research and diagnosis, *APMIS*, 96 (1988) 379–394.
16 Hashizume, K., Kanda, K. and Burke, R.E., Medial gastrocnemius motor nucleus in the rat: age-related changes in the number and size of motoneurons, *J. Comp. Neurol.*, 269 (1988) 425–430.
17 Korbo, L., Pakkenberg, B., Ladefoged, O., Gundersen, H.J.G., Arlien-Søborg, P. and Pakkenberg, H., An efficient method for estimating the total number of neurons in rat brain cortex, *J. Neurosci. Methods*, 31 (1990) 93–100.
18 Mann, M.D., Glickman, S.E. and Towe, A.L., Brain/body relations among myomorph rodents, *Brain Behav. Evol.*, 31 (1988) 111–124.
19 Martin, J.H. and Jessell, T.M., Anatomy of the somatic sensory system. In E.R. Kandel, J.H. Schwartz and T.M. Jessell (Eds.), *Principles of Neural Science*, Elsevier, New York, 1991, pp. 353–366.
20 Nathaniel, E.J.H. and Nathaniel, D.R., Fine structure of the neurons of the posterior horn in the rat spinal cord, *Anat. Rec.*, 155 (1966) 629–642.
21 Nicolopoulos-Stournaras, S. and Iles, J.F., Motor neuron columns in the lumbar spinal cord of the rat, *J. Comp. Neurol.*, 217 (1983) 75–85.
22 Nurcombe, V., Wreford, N.G. and Bertram, J.F., The use of the optical disector to estimate the total number of neurons in the developing chick lateral motor column: effects of purified growth factors, *Anat. Rec.*, 231 (1991) 416–424.
23 Pakkenberg, B. and Gundersen, H.J.G., Total number of neurons and glial cells in human brain nuclei estimated by the disector and the fractionator, *J. Microsc.*, 150 (1988) 1–20.
24 Palay, S.L. and Chan-Palay, V., General morphology of neurons and neuroglia. In E.R. Kandel (Ed.), *Handbook of Physiology*, Section 1, *The Nervous System*, Vol. I, *Cellular Biology of Neurons*, Part 1, American Physiological Society, Bethesda, MD, 1977, pp. 5–37.
25 Ren, J.Q., Aika, Y., Heizmann, C.W. and Kosaka, T., Quantitative analysis of neurons and glial cells in the rat somatosensory cortex, with special reference to GABAergic neurons and parvalbumin-containing neurons, *Exp. Brain Res.*, 92 (1992) 1–14.
26 Sakla, F.B., Postnatal growth of the cervical spinal cord of the albino mouse and the dendritic organization of its ventral horn cells, *J. Comp. Neurol.*, 113 (1959) 491–507.
27 Sakla, F.B., Post-natal growth of neuroglia cells and blood vessels of the cervical spinal cord of the albino mouse, *J. Comp. Neurol.*, 124 (1965) 189–202.
28 Schmalbruch, H., The fibre type composition of different muscles

in different species. In A. Oksche and L. Vollrath (Eds.), *Handbook of Microscopic Anatomy*, Vol. II/6, *Skeletal Muscle*, Springer, Berlin, 1985, pp. 188–195.

29 Schmidt-Nielsen, K., Warm-blooded vertebrates: what do metabolic regression equations mean?, In *Scaling. Why is animal size so important?*, Cambridge University Press, Cambridge, 1984, pp. 75–89.

30 Sterio, D.C., The unbiased estimation of number and sizes of arbitrary particles using the disector, *J. Microsc.*, 134 (1984) 127–136.

31 Todd, A.J., An electron microscope study of glycine-like immunoreactivity in laminae I–III of the spinal dorsal horn of the rat, *Neuroscience*, 39 (1990) 387–394.

32 West, M.J. and Gundersen, H.J.G., Unbiased stereological estimation of the number of neurons in the human hippocampus, *J. Comp. Neurol.*, 296 (1990) 1–22.

33 West, M.J., Slomianka, L. and Gundersen, H.J.G., Unbiased stereological estimation of the total number of neurons in the subdivisions of the rat hippocampus using the optical fractionator, *Anat. Rec.*, 231 (1991) 482–497.

34 Williams, R.W. and Herrup, K., The control of neuron number, *Annu. Rev. Neurosci.*, 11 (1988) 423–453.

35 Willis Jr., W.D. and Coggeshall, R.E., Functional organization of dorsal horn interneurons. In *Sensory Mechanisms of the Spinal Cord*, Plenum, New York, 1991, pp. 153–215.